

Ten mathematical investigations

A random sample from the mathematics atlas - September 2026

Choose an investigation whose tools you know. Make things, try ideas, and explain what you notice. You can work with a partner and use scrap paper. A good question can last longer than one meeting.

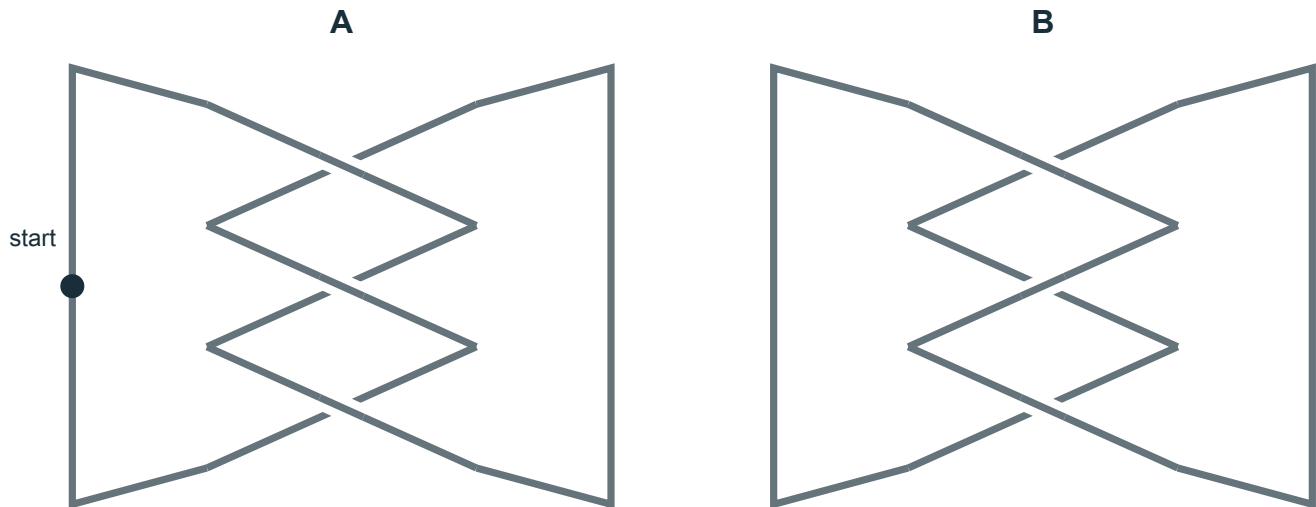
INVESTIGATION	PAGE
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Print selected pages, single-sided. The separate facilitator guide contains answers and hints. Some investigations need algebra or calculus; the tools above are prerequisites, not age labels.

Almost the same. The same knot?

Trace carefully before coloring. Each drawing is one closed cord.

A solid bridge goes OVER; a gap goes UNDER. You may bend, stretch or slide the cord. You may not open an end or pass one strand through another.



26.1-2. Follow the whole cord in each picture. What changed? Which can become a plain circle? Predict, then try closed cords if you have them.

My prediction and what I tried

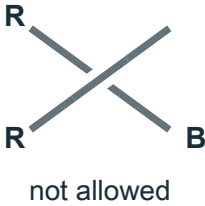
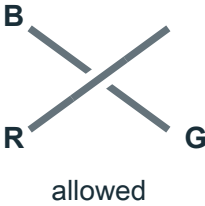
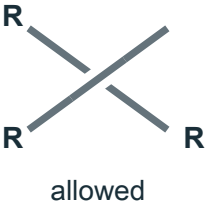
26.3. Start at the dot on A. Trace to the next UNDERPASS. Begin a new section just after it. At an OVERPASS, stay in the same section. Continue until you return to the starting section.

How many sections? Mark them on A above.

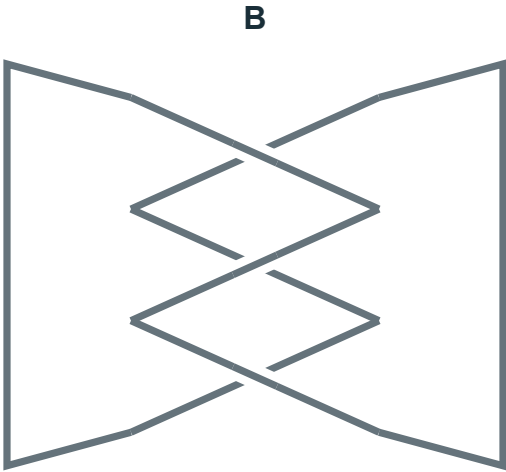
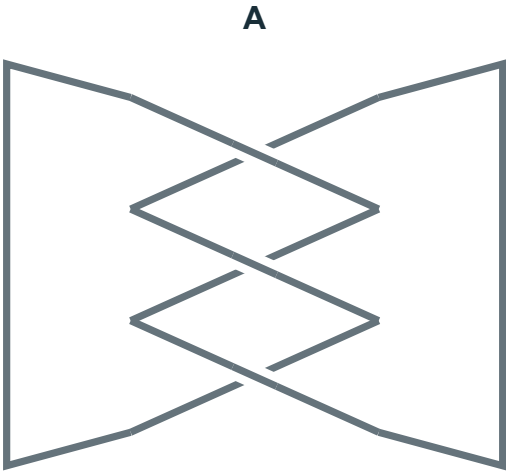
One color or three. Never two.

Use three pencils, or write R, B and G on the sections.

Color each whole section between underpasses. At a crossing, its overpass and two underpass ends must be ALL THE SAME or ALL DIFFERENT. The overpass keeps its color.



26.5. Can you color A with more than one color? Can you do the same for B? Check the outside joins, too.



26.6. For B, try two different top colors. Follow the forced colors down. Do the bottom colors match the top colors they must join?

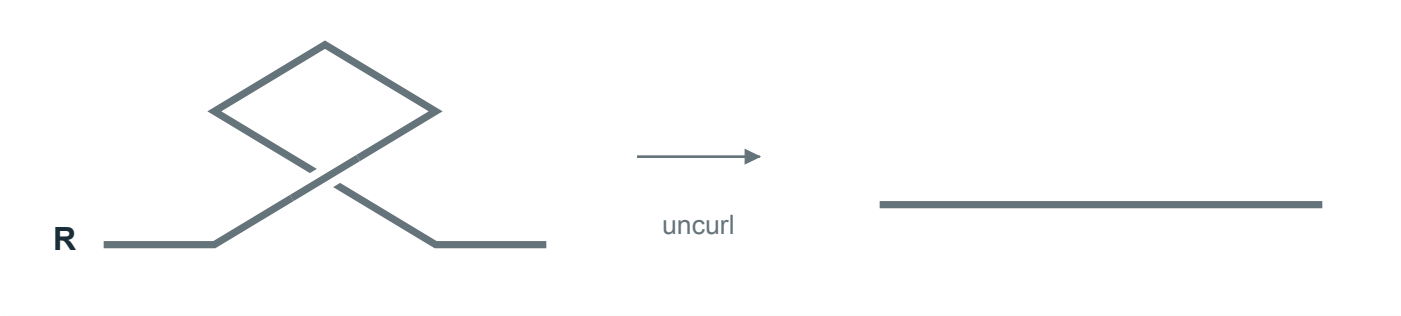
26.7. Count every legal coloring of each FIXED drawing. Color names matter.

Drawing	Total colorings	Using more than one color
A		
B		
Plain circle		

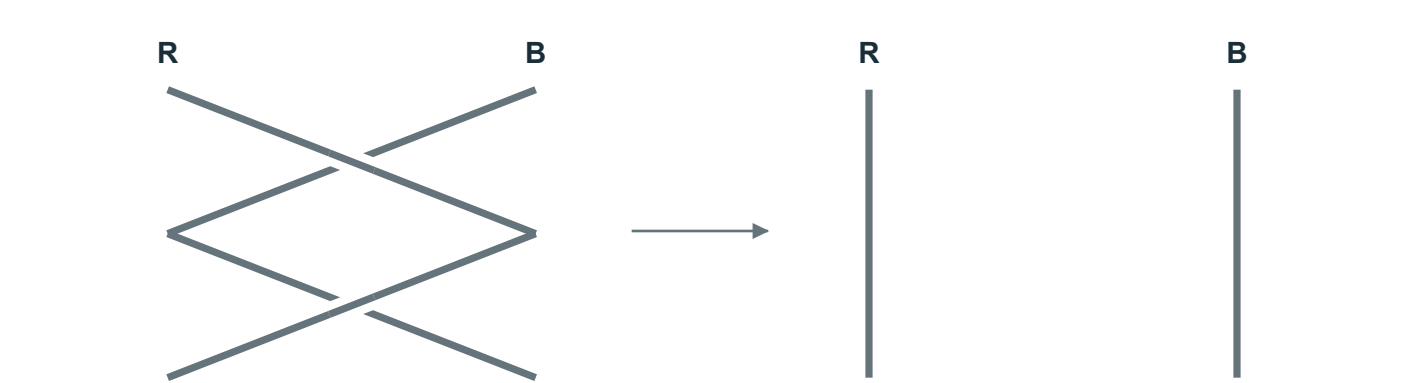
Do the colors survive a real move?

Complete every section. The gap always belongs to the underpass.

26.8. Curl or uncurl. Start red on the left. What color reaches the right? Could a different completion work?



26.9. Slide two crossings away. Start red on the left and blue on the right. Compare the bottom colors. Then try equal starting colors.



Top colors	Bottom, before slide	Bottom, after slide
R B		
R R		

26.10. Why do equal and different starting colors cover EVERY choice after renaming colors? Is each completion forced?

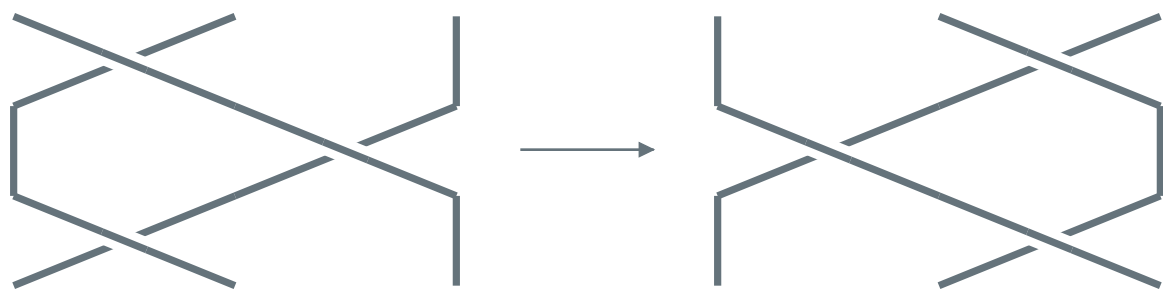
My reason

26.11 · Can B’s first two crossings slide away? Sketch what remains on the back. Why is changing one crossing a different action?

One last move. A knot certificate.

Optional: organize cases, then use a supplied fact about all deformations.

26.12. Use the SAME top colors in both pictures. Complete the sections and compare the bottom colors, left to right.



Top colors	Bottom, left picture	Bottom, right picture
R R R		
R R B		
R B R		
B R R		
R B G		

26.13. Why do these five patterns cover all starting colors after renaming? Could a section have two possible completions?

My cases and uniqueness argument

FACTS TO USE: every allowed cord deformation can be drawn as these three kinds of local move and ordinary redrawing. Every version pairs each coloring before with exactly one afterward. Our checks illustrate this fact.

26.14. Why can A, with 9 colorings, never become a circle with 3?

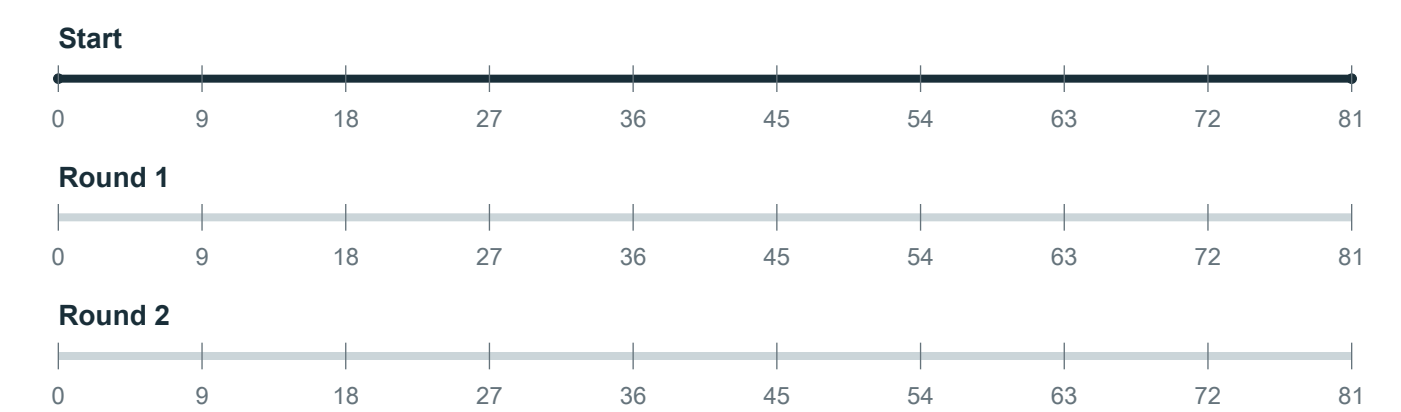
My obstruction

26.15 · Does our argument prove that matching counts mean the same knot? Why?

Which places escape the eraser?

Start with thirds and whole-number positions. Keep the boundary dots.

Start with the path from 0 to 81. Each round erases the MIDDLE THIRD of EVERY surviving piece. Keep both boundary dots of each erased gap. Draw two rounds on the blank copies.



29.2. Which places will EVER be erased? Give a first round, or a reason the place stays safe forever.

Place	First erased in round... OR safe forever	Why?
9		
15		
18		
27		
40		
54		

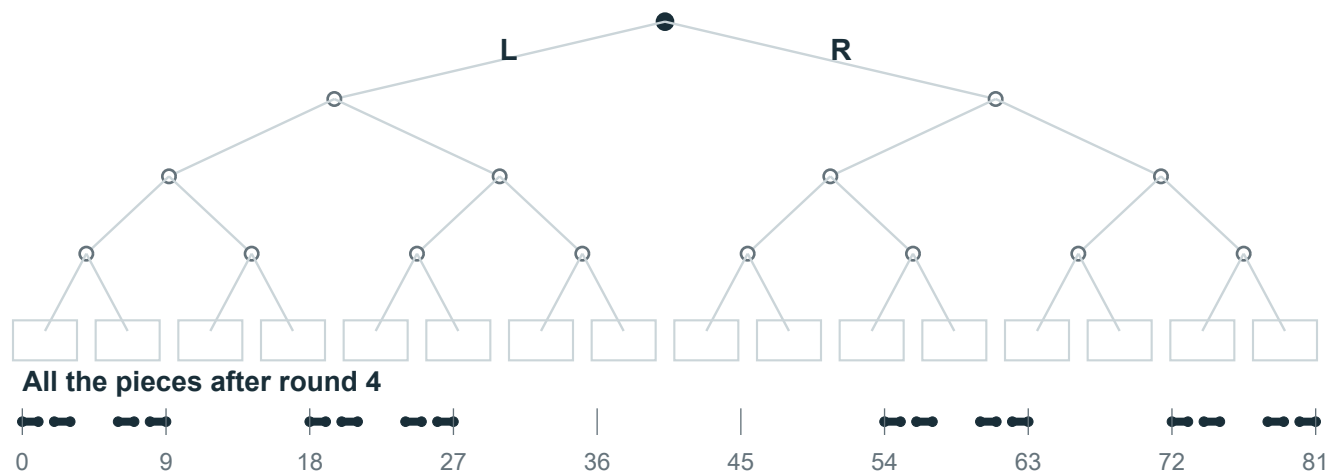
29.3. Zoom in on a surviving piece. Find a place that lasts two rounds but is erased in the third.

My place and the gap that catches it

Give every piece an address

A new representation: follow L for left, R for right.

At each split choose the left piece (L) or right piece (R). Start with the whole 0-to-81 path. Four letters name one piece after four rounds.



29.4-5. Find the endpoints for two addresses, then recover the missing address.

Four-letter address	Left endpoint	Right endpoint
LLRR		
RLRL		
	20	21

29.6. Write EVERY four-letter address beginning LR. How do you know none is missing?

29.7. Explain the number of pieces without counting the picture. Could two different addresses name the same piece?

My count and why different addresses cannot collide

Win two challenges at once

More pieces does not have to mean more total length.

29.8. Add only surviving lengths; leave out the gaps. Complete the table.

Round	Pieces	Length of one piece	Total surviving length
0	1	81	81
1			
2			
3			
4			
5			
6			

29.9. Keep MORE THAN 50 pieces with total length LESS THAN 10. Which is the first round that wins both challenges? Why can no earlier round work?

My round and certificate

29.10. Name five exact places that will NEVER be erased. Explain what protects them after a million rounds.

Why these places stay safe

The list that always misses a point

Optional infinity page: use endless addresses and careful “every row” reasoning.

FACT TO USE: an endless L/R address selects nested closed pieces shrinking to length 0. Exactly one point lies in all of them. Paper cutting alone does not prove this fact.

29.12. Why do two different endless addresses lead to different points?

Look where their addresses first differ.

29.13. Make a new address: differ from row 1 at letter 1, row 2 at letter 2, and so on. Fill six new letters.

Row	First six letters; the address continues
1	L L L L L L ...
2	R R R R R R ...
3	L R L R L R ...
4	R L R L R L ...
5	L L R R L L ...
6	R R L L R R ...

My new address begins

29.14. Six letters beat six rows, not an endless list! Give a rule for EVERY later letter. Why does your new address still name a surviving point?

A rule that works for every row n

29.15 · Two different questions: How can the survivors fit inside an arbitrarily small total length, yet defeat every list?

Length reason (continue on back)

List reason (continue on back)

Catch a claim with two routes

Core: signed fractions, symbolic algebra, and rational versus irrational numbers.

The machine accepts every rational input. Its promise is $f(x+y)=f(x)+f(y)$, and $f(1)=3$. An output is forced only if you can prove it from those promises.

G1 For each claim, mark IMPOSSIBLE or FORCED. Certify your verdict using two addition routes to the same input. Do not assume a formula for f .

$$f(0) = 1$$

Verdict: _____

Route A:

Route B:

$$f(-1) = -3$$

Verdict: _____

Route A:

Route B:

$$f(2/3) = 21/10$$

Verdict: _____

Route A:

Route B:

$$f(5/6) = 5/2$$

Verdict: _____

Route A:

Route B:

G2 Choose a new awkward rational input and a tempting wrong output. Trade with a partner: expose the contradiction using the addition promise.

Input: _____ Proposed output: _____

Certificate:

A whole infinite deck

A proof for a whole set of inputs; then one new kind of input.

You need not test every fraction separately. Try to trap them all with one argument.

- G3** Let m be any integer and n any positive integer. Explain what n copies of the input m/n add to. Use that to force $f(m/n)$. Include why your argument handles zero and negative m .

First: zero and integer inputs

Then: n copies of m/n

Therefore: every rational input

- G4** A new input key $\sqrt{2}$ appears. Which step of your fraction proof no longer applies? Can you conclude that $f(\sqrt{2})=3\sqrt{2}$, or has that value only become a guess?

- G5** Now accept exactly $a+b\sqrt{2}$ with rational a,b . You may use: each input has exactly one pair (a,b) . Invent pair addition. Test it on $(1+2\sqrt{2})+(-3+\sqrt{2}/2)$.

$(a, b) + (c, d) =$ _____

Our example: _____

Invent two machines that disagree

Full core: invent and verify two different functions on the same domain.

Both machines must accept exactly the inputs $a+b\sqrt{2}$, obey the addition promise, and send 1 to 3. You may design their insides.

- G6** Make Machine A send $\sqrt{2}$ to 0 and Machine B send $\sqrt{2}$ to 5. Give formulas in a, b . Prove each obeys addition for two arbitrary input pairs.

MACHINE A



Formula: _____

Why it obeys addition: _____

MACHINE B



Formula: _____

Why it obeys addition: _____

Addition proof for arbitrary pairs

- G7** Give an input where they disagree. Prove they agree on EVERY rational input. Why does the uniqueness of (a, b) matter?

- G8** Optional continuity challenge. For B, compare outputs at 1.4142 and $\sqrt{2}$; you may use $1.4142 < \sqrt{2} < 1.4143$. Does one close pair prove discontinuity? What happens for rational inputs arbitrarily close to $\sqrt{2}$? Which extra promise rules B out?

It reached zero. Is it parked?

Core: signed subtraction; track two cards in the right order.

Start OLD=1 and CURRENT=1. Each tick: command = $-OLD$; new position = $CURRENT + \text{command}$. Then replace BOTH cards: OLD becomes the position before the move, CURRENT becomes the new position.

OLD REPORT = 1

CURRENT POSITION = 1



- D1** Play until you think the story repeats. Record enough ticks to convince a partner. Circle every visit to zero. Does reaching zero mean staying there?



- D2** Instead use CURRENT to make the command. Start at 1 again. What happens, and exactly what did the delayed report change?

- D3** Restore the delayed rule. Choose (old,current) that makes the NEXT position zero. Find every pair that stays at zero forever. Explain the difference.

Next position is zero

Stays zero forever

What counts as the state?

A repeated reading and a repeated complete state are different.

Two robots show the same current position. Must they have the same future?

- D4** Robot A has $(\text{old}, \text{current}) = (1, 1)$. Robot B has $(\text{old}, \text{current}) = (-1, 1)$. Predict one tick for each. Find your own pair of examples showing why one current reading is not enough.

A: $(1, 1) \rightarrow$

B: $(-1, 1) \rightarrow$

- D5** Track the ordered pair $(\text{old}, \text{current})$, starting from $(1, 1)$. Draw an arrow for each tick. Stop only when the entire pair repeats. Why does that repeat prove the future repeats?

Our pair-state map: draw the states and tick arrows

- D6** Suppose the pair after a tick is (c, d) . Recover the pair before that tick. Could two different previous pairs ever merge into the same pair?

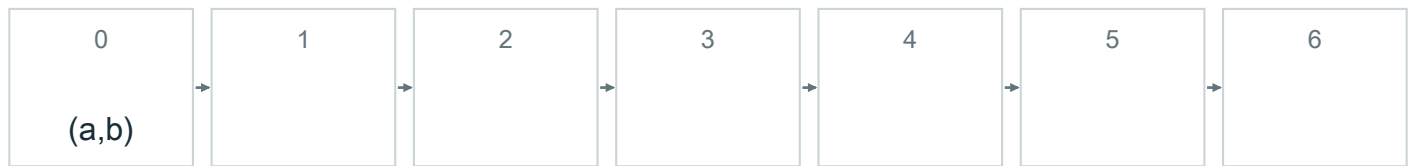
Previous pair: (_____, _____) $\rightarrow (c, d)$

Can every start come home?

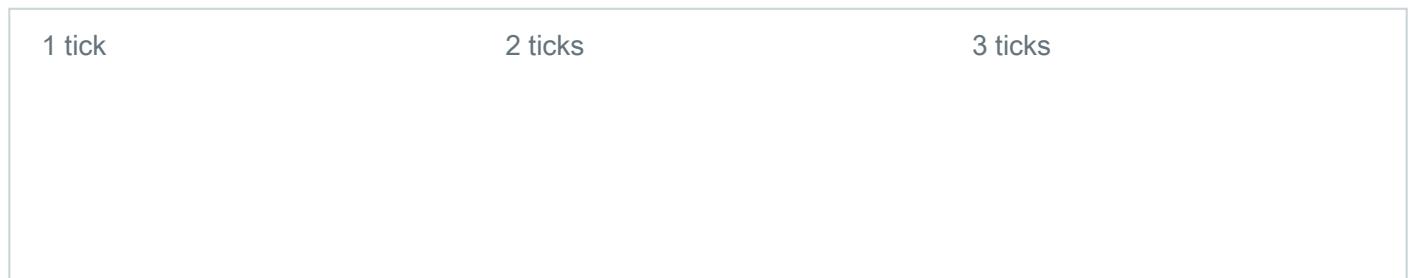
Algebra continuation: letters stand for any starting pair. Picture proof is optional.

Experiments suggest a return rule. Does it work for every starting pair, even enormous or fractional ones?

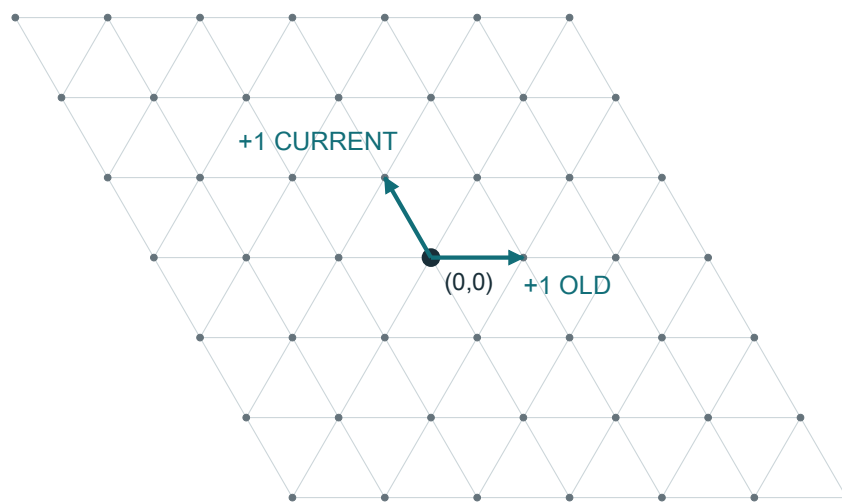
- D7** Start at (a,b) . Apply $(\text{old}, \text{current}) \rightarrow (\text{current}, \text{current} - \text{old})$ until the pair returns. Explain why your calculation works for every real a, b .



- D8** Does every nonzero pair FIRST return on the same tick? Rule out return in 1, 2, or 3 ticks. Why are those the only shorter periods you need to check?



- D9** Optional geometry. Plot the orbit of $(1,1)$, then $(1,2)$, on this triangular grid. One OLD unit points right; one CURRENT unit points up-left. What rigid motion does one tick make? Can the two unit directions prove it for all pairs?



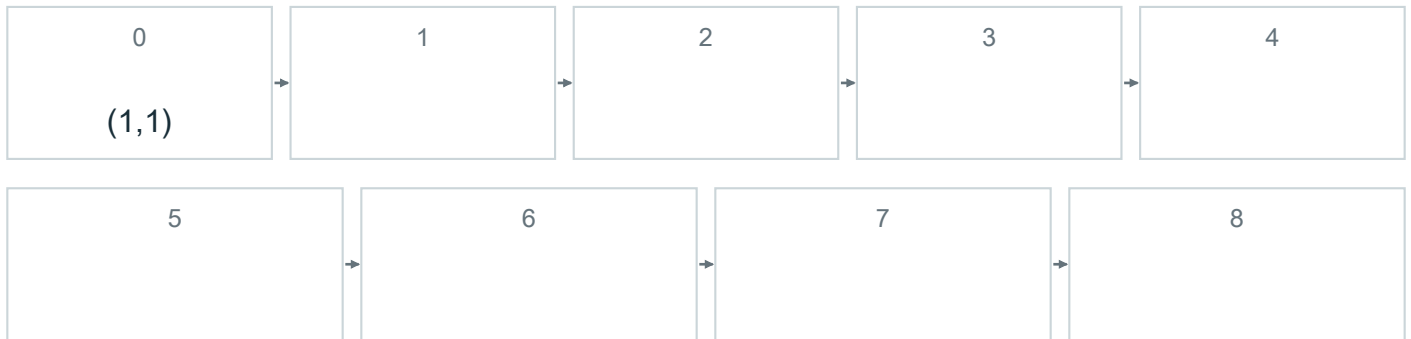
Start from the center. Follow the two arrow directions to locate a pair.

A gentler correction

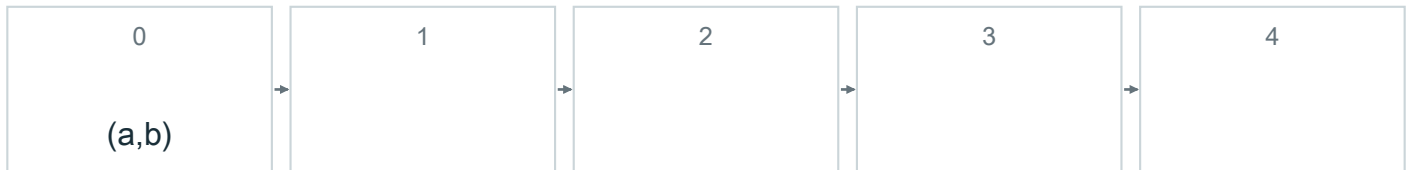
Optional extension: fractions and algebra. Plan for another session if useful.

Optional fraction-and-algebra investigation. The report is still one tick old, but the command is now HALF of its opposite. The new pair is $(\text{current}, \text{current} - \text{old}/2)$.

D10 Start at $(1,1)$. Predict whether this gentler correction helps. Run four ticks and compare the WHOLE pair with the start. Then test four more if needed.



D11 Now start at (a,b) . Work out four ticks and use the result to prove whether EVERY pair approaches $(0,0)$.



Why all four types of tick approach zero

D12 Starting at $(1,1)$, certify a tick after which BOTH cards always have magnitude at most $1/100$. It need not be the earliest. Why is a small current value alone insufficient?

A certified tick and reason

1. Choose your road

The route everyone wants

Three travelers R, G and B each choose one road. On the QUICK road, each traveler pays 1 if alone, 3 if there are two, or 5 if there are three. On the STEADY road, each always pays 4. Lower is better. All travel at the same time; these are costs, not a race.

1. Try at least three arrangements. For each, record who is on QUICK, each traveler’s cost, and the group’s total. Which arrangement would you choose as the group’s planner?

QUICK

STEADY

QUICK crowd	QUICK names	R cost	G cost	B cost	Total	Stopped?
0						
1						
2						
3						

2. Now be one traveler. Move only your counter, and only when YOUR new cost is strictly smaller. Recalculate QUICK after you join or leave! Keep moving until nobody can improve. Can you find a stopped arrangement with a different number on QUICK?

Use the four rows above. Leave “Stopped?” until you have tested the possible moves.

3. Complete the crowd map. Draw an arrow when one traveler can move to a neighboring box and strictly improve. Circle every box where play can stop.

0 on QUICK

1 on QUICK

2 on QUICK

3 on QUICK

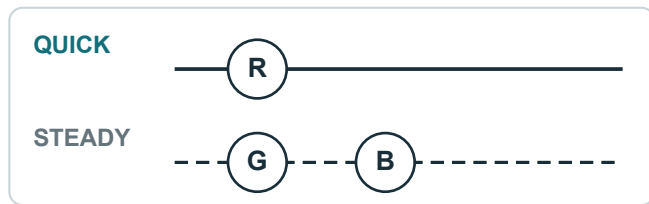
2. What changes when G switches?

The route everyone wants

Keep the same costs: QUICK 1 / 3 / 5 per traveler; STEADY 4. A stopped arrangement is called **stable**: nobody can lower their own cost by moving alone.

4. Start with R alone on QUICK, and G and B on STEADY. G switches to QUICK. Show the before-and-after costs. Who benefits? Who is worse off? What happens to the total?

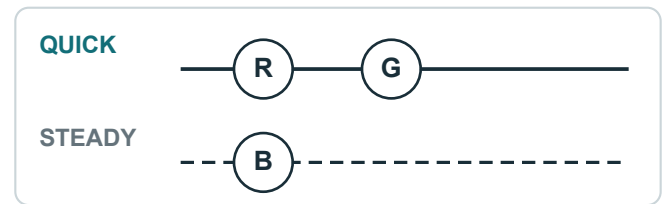
BEFORE



R: ____ G: ____ B: ____

total: ____

AFTER



R: ____ G: ____ B: ____

total: ____

5. Find the cheapest total and prove no arrangement is cheaper. Then explain why the travelers do not stay there under the strict-improvement rule. Your proof should cover every possible crowd, not just three examples.

0 on QUICK

total: ____

1 on QUICK

total: ____

2 on QUICK

total: ____

3 on QUICK

total: ____

6. The travelers are at a stable arrangement. Can one traveler's move lower the GROUP's total even though that traveler refuses to make it? Draw one such move and label every changed cost. What did "stable" fail to guarantee?



3. When “just as good” is good enough

The route everyone wants

Change STEADY’s cost to 3. QUICK still costs 1 / 3 / 5. First keep the strict-improvement rule. Then try this NEW rule: a traveler may move whenever the new cost is no larger, including a tie.

7. Under the strict rule, which crowd sizes are stable now? Under the new tie rule, can play continue forever? Act out an example.

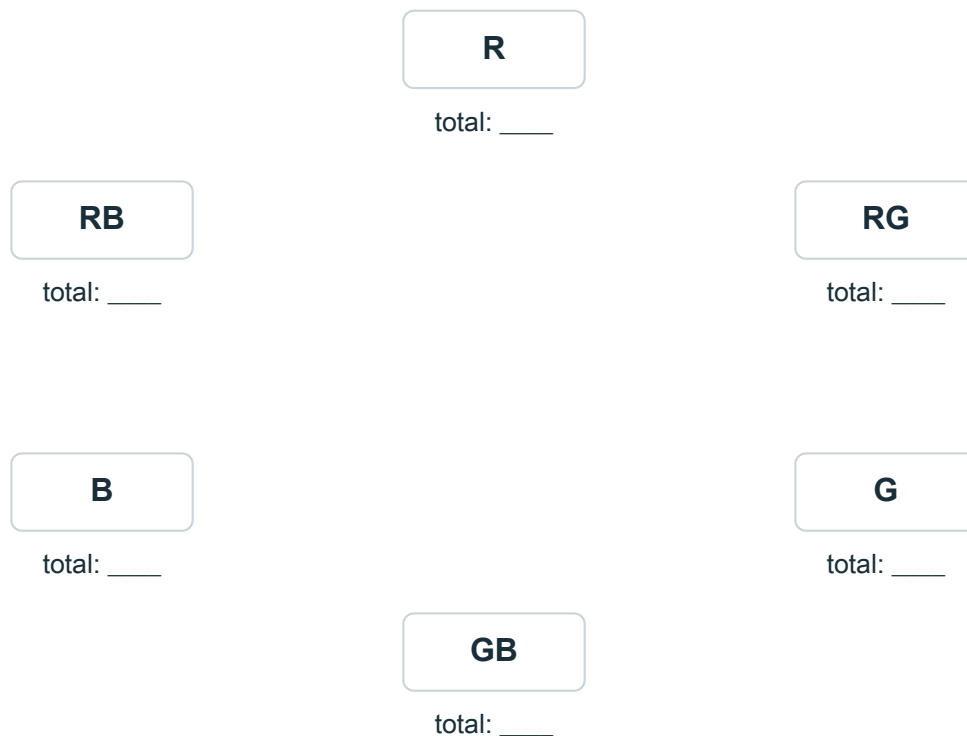
0 on QUICK

1 on QUICK

2 on QUICK

3 on QUICK

8. A box lists exactly the travelers on QUICK; everyone else is on STEADY. Join two boxes when ONE traveler can switch between them at unchanged personal cost. Put an arrow in each allowed direction. Find a loop visiting all six boxes once before returning.



9. Label each box with the group’s total cost. Around your loop, does the total always decrease? Explain how each moving traveler can avoid doing worse while the group sometimes does worse.

1. Spend now or save?

The robot's five-day battery

The robot begins day 1 with 5 battery tokens and 0 points. Each day it may take that day's job once, paying the battery cost and earning the points, or skip it. Spent battery never returns. No recharge. All five jobs are known now.

DAY 1
cost 2 battery
earn 5 points

DAY 2
cost 2 battery
earn 6 points

DAY 3
cost 3 battery
earn 9 points

DAY 4
cost 1 battery
earn 4 points

DAY 5
cost 2 battery
earn 8 points

1. Act out three different plans. Record the days you take, battery spent and points earned. What is your best score? Can your partner beat it?

Days taken	Battery spent	Points

2. A robot says, "Take each job when it comes, whenever I have enough battery." Another says, "Choose the job with the biggest point reward first, then fit in what else I can." Try both suggestions. Can you do better than either?

TAKE WHEN POSSIBLE

days: _____

battery: ____ points: ____

BIGGEST REWARD FIRST

days: _____

battery: ____ points: ____

3. Your best plan is a construction. What would convince someone that NO plan earns more? Keep your current best below; we will build a certificate on the next pages.

MY BEST PLAN / WHY I THINK IT IS BEST

2. Two different pasts, one future

The robot's five-day battery

The five jobs stay visible. We are just about to begin day 3.

4. Robot X took day 1 and skipped day 2. Robot Y skipped day 1 and took day 2. Fill in their battery and points. Which remaining jobs would you recommend to X? To Y? Explain what can be shared and what cannot.

ROBOT X

day 1 TAKE / day 2 SKIP

battery: ____ earned: ____

ROBOT Y

day 1 SKIP / day 2 TAKE

battery: ____ earned: ____

5. Make a help card for any robot beginning day 4. Only days 4 and 5 remain. Find the largest EXTRA score for each battery amount 0, 1, 2, 3, 4, 5. Explain why the answer stops rising.

battery	0	1	2	3	4	5
best extra						

6. A robot starts day 3 with 3 battery. Compare taking day 3 with skipping it. Use your help card instead of listing every complete five-day story. Does the robot's earlier score change the best recommendation?

TAKE DAY 3

points now: ____

battery before day 4: ____

best later score: ____

TOTAL: ____

SKIP DAY 3

points now: ____

battery before day 4: ____

best later score: ____

TOTAL: ____

3. A proof that works backward

The robot's five-day battery

"Best extra score" means only points from the named day onward. Once a help card is proved, anyone with that day and battery can use it. Job cards: day 1 costs 2/pays 5; day 2 costs 2/pays 6; day 3 costs 3/pays 9; day 4 costs 1/pays 4; day 5 costs 2/pays 8.

7. Complete these three day 3 help cards by comparing TAKE with SKIP. Battery 1: __. Battery 3: __. Battery 5: __. Use the day 4 card you already made. Show both possibilities whenever taking is legal.

DAY 3 / BATTERY 1

TAKE: _____

SKIP: _____

BEST: _____

DAY 3 / BATTERY 3

TAKE: _____

SKIP: _____

BEST: _____

DAY 3 / BATTERY 5

TAKE: _____

SKIP: _____

BEST: _____

8. Certify your best score. Work through decision boxes (a)-(c) in order, comparing TAKE with SKIP. Then use box (d) to trace a plan that reaches the bound.

(a) DAY 2 / BATTERY 3

TAKE: _____

SKIP: _____

BEST: _____

(b) DAY 2 / BATTERY 5

TAKE: _____

SKIP: _____

BEST: _____

(c) DAY 1 / BATTERY 5

TAKE: _____

SKIP: _____

BEST: _____

(d) TRACE YOUR PLAN

days: _____

points: ____ battery: ____

9. Change only day 5's reward from 8 points to 3. Find a new best plan. Is there more than one? What part of your old reasoning must be updated? This is an optional fresh investigation: do not assume the old plan stays best.

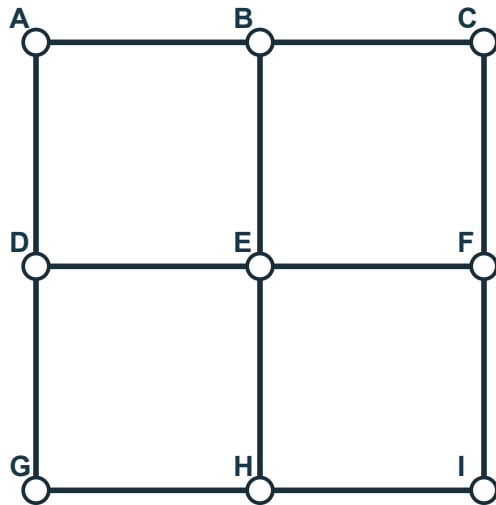
Keep the rooms. Lose the loops.

Trace routes · count to 12 · try movable road strips

Keep all nine rooms connected. Remove roads until there is no loop. A loop returns to its start without reusing a road or visiting another room twice. Going out and back on one road is not a loop.

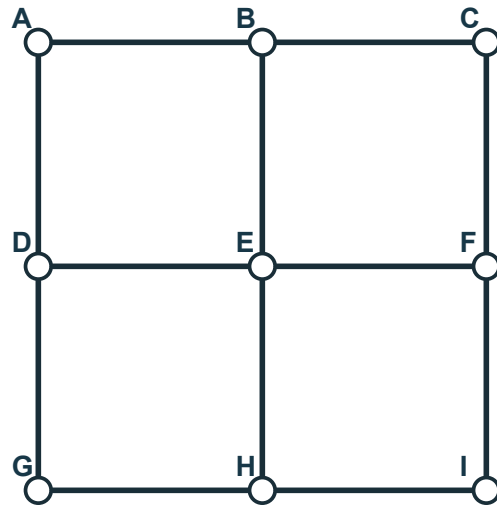
25.1-2. Make two different successful mazes. Mark every road you remove.

First maze



Roads removed: _____

Another maze



Roads removed: _____

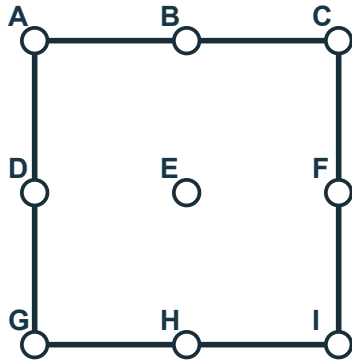
25.3. Can you succeed by removing fewer? Give your partner something to check, not only “I tried.”

My construction and my reason

A certificate for every maze

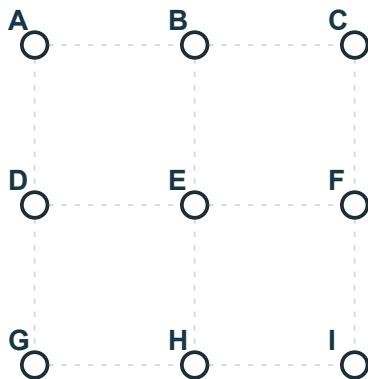
Keep your first maze nearby. Check both rules, then build a certificate.

25.4. This design has eight roads. Does it satisfy BOTH rules? Mark what goes wrong.



What fails?

25.5. Draw your kept roads over the faint guides; leave missing roads faint. Cross out a one-road room and its road. Repeat until one room remains. Record the room order.



1

2

3

4

5

6

7

8

Last room: _____

25.6. Each step loses how many rooms? How many roads? Explain how this forces the road count for every nine-room maze.

My paired room-and-road argument

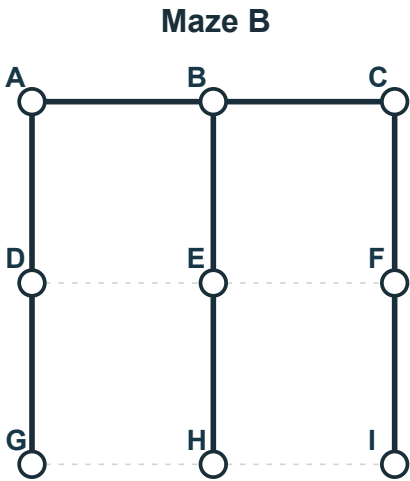
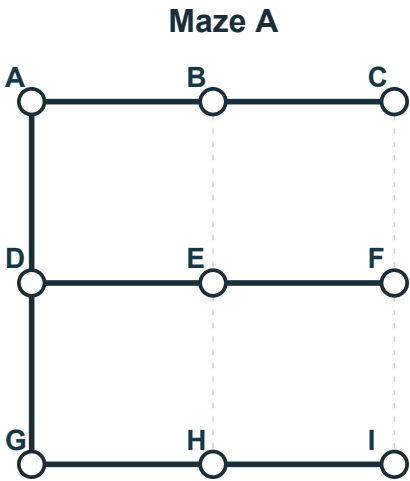
Go further (25.7): Why must another one-road room always exist? Try following a path as far as it can go.

Repair the maze one swap at a time

A road is named by its endpoints: for example, AB.

One swap: restore one missing road, then remove a different road. After each whole swap, every room must be reachable and no loop may remain. Use only the original grid roads.

25.8-9. Turn Maze A into Maze B. Find a shortest swap plan.



Swap	Restore road	Remove road
1		
2		
3		
4		
5		
6		

25.10. Why can no plan use fewer swaps?

A lower bound

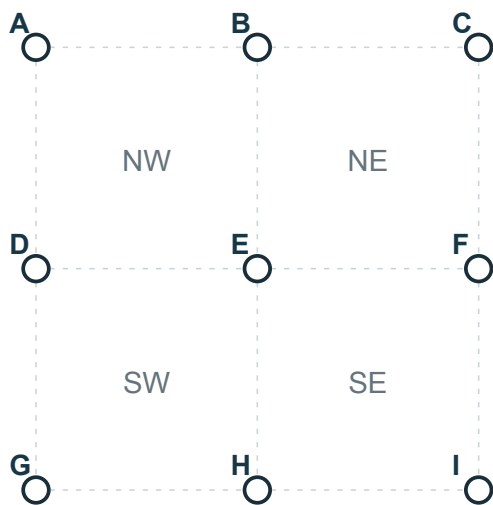
25.11 · What loop appears after you restore a road? Which road can you remove?

Turn your maze inside out

Optional: the same picture can describe a different network.

Draw your kept roads as bold WALLS over the faint guides. Missing roads stay faint: they are OPENINGS. Water crosses openings, never bold walls or wall corners. OUTSIDE wraps around all four sides.

25.12. Color where water from OUTSIDE can flow. Does it reach every courtyard?



OUTSIDE

Your maze as walls



Your map of regions

25.14. On the right, draw one link for each removed road, joining its two neighboring regions. What network appears?

I notice...

25.13. Try another maze. Can a courtyard stay trapped? Explain your prediction with a picture or words.

A reason, not only two examples

A computer walks along tangents

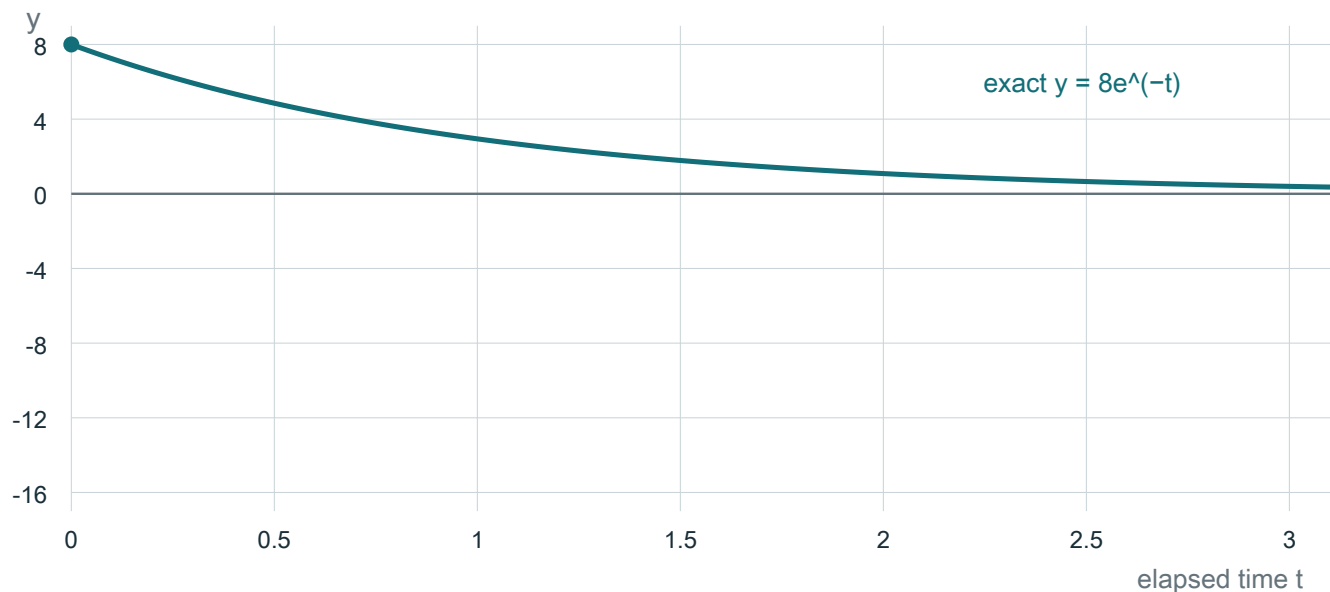
Calculus core: derivatives as slopes, exponentials, algebra. Calculator welcome.

Our model is $y' = -y$, $y(0) = 8$. Here y is the difference from room temperature, not absolute temperature. The exact model solution is $y(t) = 8e^{-t}$.

- C1** Check the supplied solution using its derivative and initial value. What must remain true about its sign and direction of change for every finite $t \geq 0$?

- C2** At (t, y) , follow slope $-y$ for a duration $h > 0$. Derive the next value. From $(0, 8)$, draw ONE step for $h = 1/2$, $3/2$, and 3 ; label endpoints by actual time. Which violate the model's behavior?

Update rule: _____



- C3** Does a negative simulated y mean negative absolute temperature? Explain its actual meaning and why it is still wrong for this model and start.

Settling down is not enough

Classify a parameter; then test a claim about accuracy.

Keep the SAME positive step size h at every update. Say that the simulator settles if its values tend to zero as the number of updates grows.

C4 Express an update as multiplication by one number. Classify ALL $h > 0$ in the table. Include every boundary value.

Multiplier: _____

Behavior	Values of h
Strictly positive, decreasing	
Hits zero in one step	
Alternates and settles	
Bounded, persistent oscillation	
Magnitude grows	

C5 Disprove: “If it settles, its answers are accurate.” Compare at the SAME elapsed time. Calculate relative error $|\text{approximate} - \text{exact}|/\text{exact}$.

h	time	approximate	exact	relative error

C6 Change the physics to $y' = -2y$. Find all positive h that make the simulator settle, and all that keep EVERY value strictly positive. Explain without a long sequence.

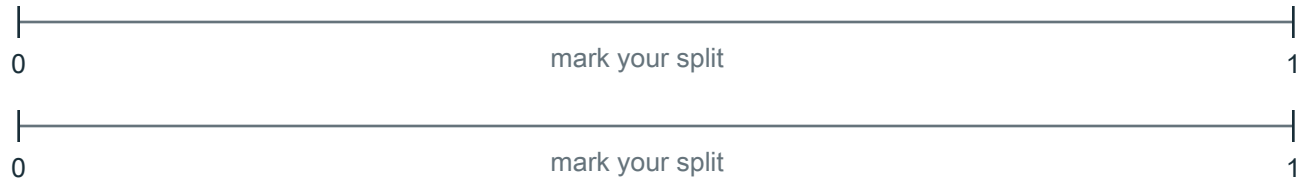
New multiplier and two parameter ranges

Spend two updates wisely

A design problem: optimize where to spend a limited number of updates.

Return to $y' = -y$. You must estimate the value at elapsed time $t=1$ using exactly TWO Euler updates. You choose the first duration h , with $0 < h < 1$. The second duration is $1-h$. Start from $y=8$.

C7 Try two split times h . Derive the final value as a function of h . Both estimates must finish at $t=1$.



Final value formula

C8 Which h gives the closest answer to $8/e$? Prove it beats EVERY two-step choice. Why does maximizing the simulated value minimize the error here?

Optimal choice and proof

C9 Replace one duration H by positive u, v with $u+v=H$. Compare the two multipliers algebraically. What improves for $0 < H \leq 1$? What exact-model comparison is needed to claim improved accuracy?

One step versus split step

An accuracy budget

Advanced continuation: choose a schedule and prove its cost is minimal.

You may choose any positive step durations adding to 1. Goal: approximate $y(1)=8/e$ with relative error at most 10%, using as few Euler updates as possible. A calculator is allowed.

- C10** Find a schedule meeting the target. Equal steps are worth trying, but unequal ones are allowed. Give durations, estimate, and relative error.

Schedule / final estimate / relative error

- C11** Prove fewer updates than your proposed minimum cannot meet the target. You may use: among nonnegative factors with a fixed sum, the equal factors give the largest product. Identify the factors and sum; include schedules with still fewer updates.

A lower bound for every shorter schedule

- C12** Optional repair: choose the END slope, so $y_{\text{new}}=y_{\text{old}}-h \cdot y_{\text{new}}$. Solve for y_{new} . Is it positive and settling for every $h>0$? Does that buy 10% accuracy with one step to $t=1$?

New update rule, behavior, and accuracy test

1. Which shore?

A fair price for a wandering token

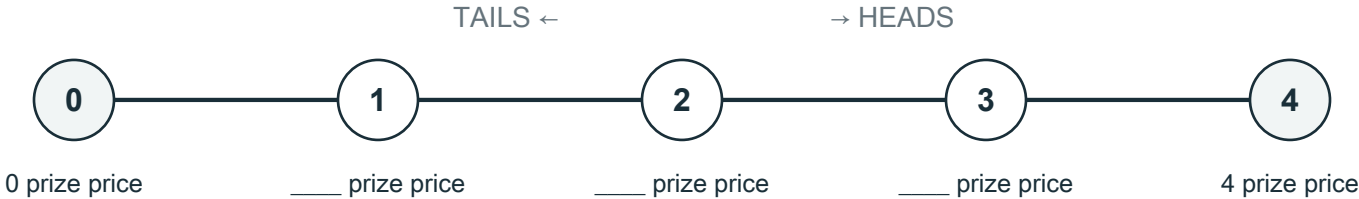
Use a fair coin with independent tosses. Choose a start at 1, 2 or 3 on the track 0-1-2-3-4. Heads moves one step right; tails one step left. Stop at shore 0 or shore 4. Reaching 4 wins 4 prize counters; reaching 0 wins none. There is no prize for a long walk.



1. Try the walk from two different starting places. Record at most four finished trips. Before more tossing, predict: which start gives the best chance of reaching 4? Can any of those starts guarantee a win?

Start	A short path / toss record	Shore

2. Imagine buying a token already standing at 1, 2 or 3. A fair price equals its average eventual prize over many independent games. Guess a fair price for each start. Record a reason, not just a fraction from your four trips.



3. Two students disagree: "Start 2 must win because it is in the middle." "Start 2 wins about half the time because left and right paths can be paired." Which argument makes sense? Explain what you think happens.

2. Build a chance skyline

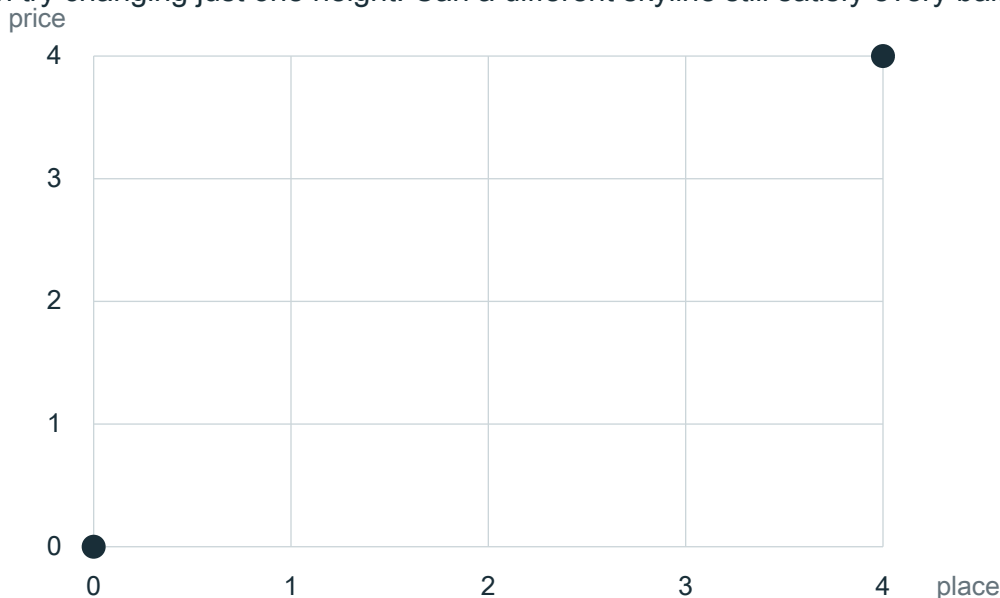
A fair price for a wandering token

The endpoints have prices 0 and 4. At any interior place, one fair toss sends the token to one of two neighbors. Use that first toss to reason about the price HERE before you know the whole path.

4. If the two possible next places have prices a and b , what fair price should HERE have? Explain with two equally likely possibilities. Use words, a balance drawing or an equation.



5. Choose heights at 1, 2, 3 so every interior height balances its two neighboring heights. Draw your skyline. Then try changing just one height. Can a different skyline still satisfy every balance?



6. Give a reason that forces your heights. What does “middle height equals the average of its neighbors” say about the two gaps beside it? Finally turn your prices into chances of winning 4 counters.

gap 1: ____ gap 2: ____ gap 3: ____ gap 4: ____

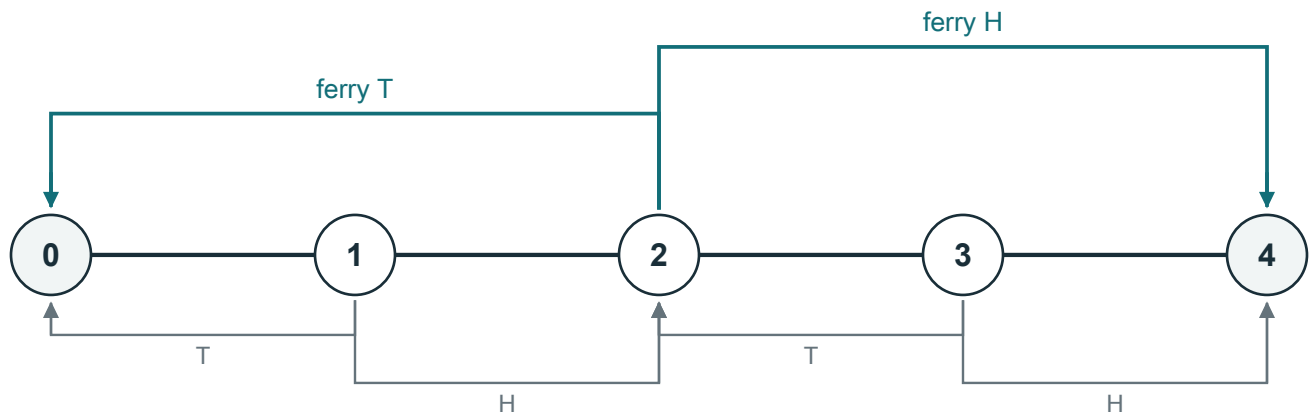
chance from 1: ____ chance from 2: ____ chance from 3: ____

3. A ferry at the middle dock

A fair price for a wandering token

Change only the move from 2. At 2, heads now ferries the token straight to 4; tails ferries it straight to 0. From 1 and 3, keep the old one-step rule. The coin is still fair and independent. Stop at either shore as before.

7. Before playing, predict which starting places have a changed chance of winning. Draw new fair prices on the ferry map. Check the first-step balance at EVERY interior place, including the ferry dock.



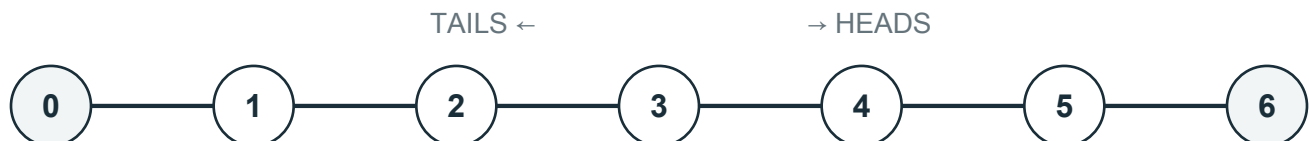
price at 1: ____

price at 2: ____

price at 3: ____

8. What changed: the possible paths, the winning chances, or both? Does having a shorter route to a shore automatically mean a better chance of reaching 4? Give evidence from this game.

9. Design a fair shortcut on 0-1-2-3-4-5-6. Keep prize 6 at the right shore and 0 at the left. Ordinary one-step moves have prices 0, 1, 2, 3, 4, 5, 6. Change the TWO destinations from one interior place so its old price still balances them. Show your arrows and explain. Both destinations must differ from the starting place.



1. No spaces allowed

Send it with fewer bits

A sender uses only 0 and 1. The receiver sees one long string, with no spaces, pauses or punctuation. Our four pictures are A $\triangle$, B $\circ$, C $\square$ and D $\star$. The day's message strip is A B A C A B D A. Spaces on this picture strip are for the sender only.

1. First use A=00, B=01, C=10, D=11. Send the strip to a partner. The receiver marks the boundaries and restores the pictures. How many bits did you send?

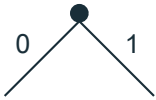
 A	 B	 A	 C	 A	 B	 D	 A

Receiver pictures: _____

2. A sender proposes A=0, B=01, C=10, D=11. Can the string 010 mean two different picture messages? Show both readings. What went wrong?

0 1 0 pictures: _____	0 1 0 pictures: _____
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3. Build a shorter reliable code for the original eight-picture strip. Use a branching tree: each left edge sends 0, each right edge sends 1, and put each picture at a LEAF, where that path ends. A picture may not sit partway along another picture's path. Try to beat 16 bits.



A = _____

B = _____

C = _____

D = _____

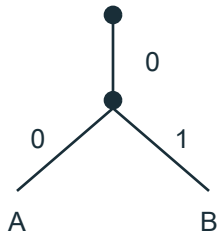
TOTAL BITS: _____

2. How do you know it is best?

Send it with fewer bits

Use the original counts: A appears 4 times, B 2 times, C 1 time and D 1 time. A picture's code length is its number of edges from the root. A tree is full when every branching point has two children.

4. Look at a tree with a one-child branching point. Can removing that unnecessary step ever make any message longer or confuse two leaves? Explain why a best tree can be full.



SHORTEN IT HERE

5. Find the full binary tree shapes with exactly four leaves. Ignore left/right mirror flips. At the root, how many leaves can go on each side? Draw the possibilities; then decide which leaf should get each picture.



root split: ____ + ____



root split: ____ + ____

6. Compute the best total bits for each shape. Explain why moving a more frequent picture to a shorter path cannot hurt. Use that fact and your shape list to prove that your best code cannot be beaten by a prefix code.

Shape 1 total: _____

Shape 2 total: _____

3. A new day, a different best tree

Send it with fewer bits

Keep four pictures and binary prefix codes. Only the frequencies change.

7. New message strip: A B C D A B C D. Which of your two tree shapes wins now? Could yesterday's shortest-total code be worse today? Show the bit totals.



balanced total: _____

long-branch total: _____

8. Invent a SIX-picture strip using every picture at least once, for which the two tree shapes have the SAME best total length. Show your strip, a best code of each shape and both totals.



BALANCED

A = _____ B = _____

C = _____ D = _____

total bits: _____

LONG BRANCH

A = _____ B = _____

C = _____ D = _____

total bits: _____

9. Trade codes and invent a difficult test message for your partner. Send only the bits. The receiver must decode it without guessing. Explain why stopping at a leaf and returning to the root always works for your code.

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